

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7649
- **Team Leader:** N.Sopya

2. FUNCTIONAL REQUIREMENTS Functional requirements define the specific behaviors and capabilities the ComicCraft system must possess.

- **FR1 - User Input Handling:** The system must provide an accessible HTML web interface with input fields for Story Prompt, Character Name, Setting, Tone, and Art Style.
- **FR2 - Story Generation:** The system must integrate securely with Google Gemini (Flash & Pro) to dynamically generate a structured 5-panel story outline and narrative dialogue.
- **FR3 - Image Generation:** The system must integrate with Hugging Face Diffusers (Stable Diffusion v1.5) to automatically generate high-quality illustrations matching the AI-created scene descriptions.
- **FR4 - Dynamic Layout Rendering:** The backend must parse the generated text and images, rendering them iteratively onto a preview HTML page using Jinja2 templates.
- **FR5 - PDF Compilation:** The system must allow users to export the generated comic. The backend must use the FPDF library to structure images and text onto consecutive pages efficiently.
- **FR6 - Fallback Mechanism:** The system must implement robust error handling (e.g., dual-engine switching or generating text-based fallback images) to ensure the application does not crash during external API latency or HTTP 402/500 errors.

3. NON-FUNCTIONAL REQUIREMENTS Non-functional requirements specify the quality attributes and operational constraints of the system.

- **NFR1 - Performance:** The entire generation process (text and multiple images) should optimally complete within an acceptable timeframe (45-60 seconds) to maintain user engagement.
- **NFR2 - Usability:** The frontend must be highly responsive, aesthetic, and intuitive, requiring absolute zero technical or artistic expertise from the end-user.
- **NFR3 - Scalability:** The FastAPI backend must utilize asynchronous routing to ensure it is capable of handling concurrent requests for future deployment on cloud servers.
- **NFR4 - Security:** Environment variables, specifically sensitive API Keys for Gemini and Hugging Face, must be secured via `.env` files and must never be hardcoded into the source code repository.